A Baer-subplane construction of Geometry 3.3 and a correction to its printed incidence rule

L100 finite construction component

We use the fixed 22-page Halle preprint of Gottschalk and Leemans [1], physical page 19. Its later journal edition has not been collated. Put $P = \mathrm{PG}(2,4)$ and $G = \mathrm{PSL}_3(4)$. The coordinate plane, its exact identification with the retained 21-point action, and the complete coset representative numbered 10 are acquired from the accepted projective-construction, palette and source-join components. The latter representative is Geometry 3.3.

Theorem 1. *For each of the three G-orbits of Baer subplanes, the three incidence rules printed for Geometry 3.3 admit a two-object flag which does not extend to a chamber. Retain the first two rules and restrict the third to pairs having a common incident Baer subplane in the selected orbit. Each of the three resulting incidence systems is isomorphic, with its specified G-action transported by an explicit automorphism of G, to the acquired Geometry 3.3. The printed and corrected line-pair to point-pair degrees are respectively* 18 *and* 6.

This corrects an ancillary construction recipe. It does not refute the paper's principal classification, which already contains the corrected geometry. Other constructions and the complete source disposition are outside this component.

1 Complete subplane input

Lemma 1. *A seven-point subset B of P is an embedded plane of order two if and only if every ambient line meets it in one or three points. There are exactly* 360 *such subsets.*

Proof. Under the intersection condition every pair of points lies on a unique three-point intersection. These give $\binom{7}{2}/\binom{3}{2} = 7$ lines, three through each point. Each such line meets the six others, twice at each of its three points, since two lines cannot share two points. Thus the induced incidence system is an embedded Fano plane. Conversely, a Fano subplane has seven ambient lines meeting it in three points. Any other ambient line meets it in at most one point. It cannot avoid the subplane: its intersections with the seven subplane lines would be distinct, although it has only five points.

An ordered projective frame is a sequence of four points with no three collinear. There are $21 \cdot 20 \cdot 16 \cdot 9 = 60480$ ambient frames. Every frame lies in a unique Fano subplane. Indeed a projectivity sends it to $[1,0,0], [0,1,0], [0,0,1], [1,1,1]$. The three remaining Fano points are forced by intersections of the joins of these four points; they are $[1,1,0], [1,0,1], [0,1,1]$. These seven points do form a subplane, so existence and uniqueness both hold. Each Fano plane has $7 \cdot 6 \cdot 4 \cdot 1 = 168$ ordered frames. Counting frames with their unique subplane gives $60480/168 = 360$. □

The preserved first finite pass examined all $\binom{21}{7} = 116280$ subsets and saved the 360 qualifying sets before failing its later, uncorrected incidence join. The current checker uses these saved sets: it verifies distinctness and the lemma's criterion, then closes each listed orbit under the actual acquired generators. The result is three orbits of size 120. Completeness follows from the lemma; it is not inferred from a printed orbit count. No subset search is repeated.

2 The printed rule and its repair

For one orbit $\mathcal{B}$, the types are $\mathcal{B}$, unordered pairs of distinct lines, and unordered pairs of distinct points. Write pq for the line joining p and q. The printed rules are

$$B * \{\ell, m\} \iff B \cap (\ell \cup m) = \ell \cap m, \tag{A}$$

$$B * \{p, q\} \iff B \cap \{p, q\} = \varnothing \text{ and } |B \cap pq| = 3, \tag{B}$$

$$\{\ell, m\} * \{p, q\} \iff (\ell \cup m) \cap \{p, q\} = \varnothing \text{ and } \ell \cap m \cap pq \neq \varnothing. \tag{C}$$

These are literal transcriptions of the native source, not of its OCR. Let $o = \ell \cap m$. In (C), pq must be one of the three other lines through o. On each there are four points outside $\ell \cup m$. Consequently (C) has degree $3\binom{4}{2} = 18$ at every line pair.

Here is an explicit finite obstruction to flag extension. Label the retained points $0, \ldots, 20$ as in the acquired coordinate-to-permutation map, and encode a point set X by $\sum_{x \in X} 2^x$. Fix $\ell = \{0, 1, 2, 3, 4\}$ and $m = \{0, 5, 6, 7, 8\}$. In the three orbits ordered by their least encoded subplane, the following table gives a printed incident point pair and the complete two sets of Baer neighbours.

Orbit	Point pair	Neighbours of $\{\ell, m\}$	Neighbours of the point pair
0	$\{9, 11\}$	710657, 730625, 1361409, 1390593	46093, 87059, 398529, 1578273
1	$\{9, 10\}$	816129, 887297, 1206785, 1283073	47297, 88353, 399373, 1579027
2	$\{9, 10\}$	438785, 478209, 1616897, 1659393	30741, 104459, 792737, 1186113

The joining line in each case is $\{0, 9, 10, 11, 12\}$, so (C) holds. The two neighbour sets in each row are disjoint. The checker obtains each list by applying (A) and (B) to every member of that orbit, using an independent set formulation. Thus each displayed pair is a flag with empty remaining-type residue. This proves the stated defect of the printed recipe.

The repaired relation is

$$\{\ell, m\} *' \{p, q\} \iff \text{(C) holds and there exists } B \in \mathcal{B} \text{ satisfying (A) and (B).}$$

Rules (A) and (B) are unchanged. The selected orbit is essential. For additional exact context, the 3780 printed pairs split into three sets of 1260: in each pair precisely one orbit supplies common Baer neighbours, and it supplies two. The checker verifies this finite partition directly. In each repaired system the degree is six.

3 Literal positive identifications

The acquired catalogue representative has maximal parabolics H_i and objects the right cosets $H_i g$. For each of the three repaired systems the certificate supplies a type permutation, base objects x_i and a point permutation h. It checks that h preserves the actual lines and conjugates the full, explicit 20160-element group onto itself. With $\alpha(g) = h^{-1} g h$, the object map is

$$H_i g \longmapsto x_i^{\alpha(g)}.$$

The literal parabolic generators fix x_i. Moreover the verifier checks all 60480 coset-member assignments in each map, so it verifies well-definedness independently of the choice of coset representative. It then checks each typewise bijection, all 189000 directed cross-type object pairs, and all 1080 generator-object action pairs. Line pairs are pairs of five-point sets acted on by actual point permutations; no point-label/line-label identification is assumed.

All three maps pass. Therefore incidence, nonincidence and the specified action agree exactly. The already proved geometry, connectedness, residual properties, flag transitivity and RWPRI properties transport through these maps. In particular the three acquired local two-transitivity flags are all true. Equivalently, composing one map with the inverse of another gives a full incidence equivalence between any two orbit choices; this is not inferred from their equal sizes.

The certificate's finite verification, its exact acquired input interfaces and the short lemmas above constitute the proof. The acquired group order and full coset catalogue retain their previously stated computational and classification premises. The positive maps do not require a new group, subgroup, automorphism or graph-isomorphism search. The first failed join, the prelaunch ordering correction and the changed successful join are preserved separately. This component makes no novelty, measurement, open-question or whole-source claim.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL$(3, 4)$, retained Halle preprint 99–30 (1999), physical page 19. The later journal edition is not collated here.